Advanced Charging Station for EV's with Partial Power Processing

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Abstract: This study provides a comprehensive performance evaluation of an Extreme Fast Charging (XFC) station's power delivery system for Electric Vehicles (EVs), utilizing Partial Power Processing (PPP) and an Artificial Neural Network (ANN) controller. The ANN controller enables adaptive power flow management, enhancing charging efficiency and facilitating better integration with the power grid. Through detailed simulations and experimental testing, the system is assessed across various operational scenarios to determine its robustness and performance. Critical metrics such as charging duration, power quality, and grid stability are examined under different load conditions and grid disruptions. The results highlight the proposed system's ability to deliver fast and efficient charging while upholding high standards of power quality and grid reliability. This research supports the advancement of XFC station technologies, offering valuable insights into the application of intelligent control strategies to improve the effectiveness and dependability of EV charging infrastructure.

Keywords: Extreme Fast Charging (XFC) Station's, Electric Vehicles (EVs), Partial Power Processing (PPP), Artificial Neural Network (ANN) Controller, Power Quality, Grid Stability.

I. INTRODUCTION

The increasing adoption of Electric Vehicles (EVs) has necessitated the development of efficient, intelligent, and scalable charging infrastructures. Traditional EV charging systems often suffer from energy inefficiencies, high processing losses, and limited adaptability to dynamic grid and vehicle conditions. This project proposes the design and implementation of an Advanced Charging Station featuring Partial Power Processing (PPP) and an Artificial Neural Network (ANN)-based controller, aimed at optimizing power flow, reducing conversion losses, and improving overall charging efficiency. The concept of Partial Power Processing enables only a fraction of the total power to be processed through active converters, significantly reducing switching and conduction losses. This not only enhances system efficiency but also minimizes the size and cost of power electronics. To further improve the performance and adaptability of the charging system, an ANN controller is integrated to dynamically manage charging parameters based on real-time data, such as battery state-of-charge (SoC), grid conditions, and user defined constraints.

This intelligent system offers a flexible and energy-efficient solution for future EV charging infrastructures, supporting both

grid stability and user convenience. The proposed architecture combines power electronics innovation with machine learning control strategies, paving the way for next-generation smart charging stations.

Such as a building-based micro grid. In order to reduce stress on the grid infrastructure and to avoid excess demand charges, centralized energy storage, and on-site energy generation can be integrated to the common dc link. The power rating of the CHB and SST units can be significantly reduced with such local energy storage solutions.

The novelty of the proposed scheme lies in the use of dc–dc converters that employ partial power processing for each charging point as against the full power dc–dc converter-based charging solution reported in the literature. The charging units are rated only to handle a fraction of the power required for battery charging. This approach can potentially reduce the installation and operational costs in an XFC station and simultaneously improve the overall system efficiency.

II. LITERATURE SURVEY

Dr. D. Kumara Swamy, O. Sony, A. Anil Kumar, D. Prem Kumar, D. Prashanth, K. Shiva Prasad et al. This article proposes an approach for realizing the power delivery scheme for an extreme fast charging (XFC) station that is meant to simultaneously charge multiple electric vehicles (EVs). A cascaded H-bridge converter is utilized to directly interface with the medium voltage grid while dual-active-bridge based soft-switched solid state transformers are used to achieve galvanic isolation. The proposed approach eliminates redundant power conversion by making use of partial power rated dc–dc converters to charge the individual EVs. Partial power processing enables independent charging control over each EV, while processing only a fraction of the total battery charging power. Practical implementation schemes for the partial power charger unit are analyzed. A phase-shifted full-bridge converter-based charger is proposed. Design and control considerations for enabling multiple charging points are elucidated. Experimental results from a downscaled laboratory test-bed are provided to validate the control aspects, functionality, and effectiveness of the proposed XFC station power delivery scheme. With a downscaled partial power converter that is rated to handle only 27% of the battery power, an efficiency improvement of 0.6% at full-load and 1.6% at 50% load is demonstrated [1].

V. M. Iyer, S. Gulur, G. Gohil and S. Bhattacharya et al. This paper introduces a power delivery architecture for an Extreme Fast Charging (XFC) station that is meant to simultaneously charge multiple electric vehicles (EVs) with a 300-mile range battery pack in about 15 minutes. The proposed approach can considerably improve overall system efficiency as it eliminates redundant power conversion by making use of partial power rated dc-dc converters to charge the individual EVs as opposed to a traditional fast charging station structure based on full rated dedicated charging converters. Partial power processing enables independent charging control over each EV, while processing only a fraction of the total battery

charging power. Energy storage (ES) and renewable energy systems such as photovoltaic (PV) arrays can be easily incorporated in the versatile XFC station architecture to minimize the grid impacts due to multi-mega watt charging. A control strategy is discussed for the proposed XFC station. Experimental results from a scaled down laboratory prototype are provided to validate the functionality, feasibility and cost-effectiveness of the proposed XFC station power architecture [2].

Anzola, J.; Aizpuru, I.; Arruti et al. This paper focuses on the design of a charging unit for an electric vehicle fast charging station. With this purpose, in first place, different solutions that exist for fast charging stations are described through a brief introduction. Then, partial power processing architectures are introduced and proposed as attractive strategies to improve the performance of this type of applications. Furthermore, through a series of simulations, it is observed that partial power processing based converters obtain reduced processed power ratio and efficiency results compared to conventional full power converters. So, with the aim of verifying the conclusions obtained through the simulations, two downscaled prototypes are assembled and tested. Finally, it is concluded that, in case galvanic isolation is not required for the charging unit converter, partial power converters are smaller and more efficient alternatives than conventional full power converters [3].

Challa Ramaiah, Edara Venkata Chandra Sekhara Rao, Satish Kumar Peddapelli et al. Charging the EV's battery packs is a crucial task as it needs DC voltage stability and reduced circulating currents. Conventional converters (Boost or Buck-Boost) have high disturbances, ripple, and lower efficiency. These converters are unsuitable for high-current applications such as electric vehicle battery charging. In this paper, fast charging of electric vehicle batteries is done using PPCC topology. The PPCC topology comprises a high-frequency inverter, HFTF and uncontrolled rectifier for the power transfer from the source to the battery. The PPCC topology has very little ripple in the output DC voltage, even during heavy power transfer conditions, which keeps the battery pack safe. Charging from conventional grid and fossil fuel power sources is not recommendable as EVs are considered zero-emission vehicles. The batteries of EVs need to be charged by renewable sources, which generate power using natural sources like solar radiation, wind, biogas, etc. A standalone renewable source module is modelled for the PPCC charging station, which comprises solar panels and a PMSG wind source. MPPT-based PI voltage control is adopted for voltage stability in renewable sources. The boost converter of the solar panel and wind generator sources is controlled by the proposed voltage controller, supplying maximum power at stable voltages. An analysis is done on the proposed standalone renewable source PPCC topology with different operating conditions using Simulink modeling of MATLAB software [4].

Darla Nagasai Krishna, Vasupalli Manoj, Ganta Yamini et al. Charging stations are essential as the number of electric vehicles

(EVs) continues to rise. Emerging extreme fast charging (XFC) technology could allow gasoline-like recharge. This article analyzes the state-of-the-art EV charge stations and XFC technologies needed to serve current and future EV refueling needs. We explore XFC station design and popular power electronics converter topologies. We examine the advantages of adopting solid-state transformers (SSTs) in XFC stations to replace standard line-frequency transformers and assess medium-voltage SST designs for XFC.This paper provides a power distribution technique for a station that can charge numerous electric automobiles at once (EVs) (EVs). A cascaded H-bridge (CHB) converter interacts directly with the MV grid, while dual active bridge (DAB)-based soft-switched solid-state transformers (SST) maintain galvanic isolation. Using partial power DC-DC converters to charge EVs reduces wasteful power conversion. Partial power processing allows each EV to charge separately by processing a part of the battery charging power. Explores feasible partial power charger implementation techniques. PSFB charger is advised. The architecture of several charging stations and their control are examined. Experimental results from a downscaled laboratory test-bed verify the proposed XFC station power delivery method. A partial power converter rated to take just 27% of the battery power increases efficiency by 0.6% at full-load and 1.6% at 50% load [5].

Nagaraju Budidha,Mysa Aparna,Purna Chander Rao Perala, Porika Srilatha, Pujari Arun et al. This paper presents the design and implementation of an extreme fast charging station for electric vehicles (EVs) that utilizes partial power processing technology. As the demand for electric vehicles continues to rise, the need for efficient and rapid charging solutions has become critical. Traditional charging stations often struggle to meet the rapid charging requirements of modern EVs, leading to longer wait times for users. The proposed station employs partial power processing to optimize energy conversion and reduce the overall system complexity, enabling faster charging without compromising safety or performance. By selectively processing only the necessary power levels during charging cycles, the station achieves higher efficiency and lower heat generation, contributing to improved reliability and longevity. Simulation results and experimental validation demonstrate the effectiveness of the design, showcasing significant reductions in charging time while maintaining compatibility with a range of EV models. This innovative approach represents a significant advancement in EV charging infrastructure, addressing both user convenience and the evolving needs of the electric vehicle market [6].

III. METHODOLOGY

Solid State Transformer: A solid-state transformer (SST), also known as a power electronic transformer (PET), is an advanced AC-to-AC converter that replaces conventional utility-frequency transformers. Operating at high frequencies, SSTs are more compact and efficient, offering electrical isolation and voltage regulation through integrated power electronics. They can perform AC-to-DC and DC-to-AC conversions and support single- or three-

phase power, with some models handling direct DC input/output for enhanced efficiency. Due to their smaller size and lightweight design, SSTs are ideal for critical applications like data centers, electric traction systems, MVDC ships, aircraft power systems, and underground substations where space is limited.

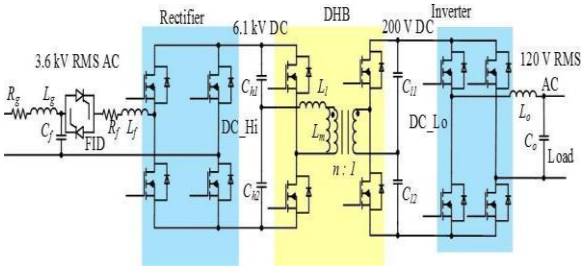


Fig. 1: SST Topology

Dual Active Bridge Converter: A Dual Active Bridge is a DC/DC converter offering galvanic isolation and bidirectional power flow. The galvanic isolation is ensured by a high-frequency intermediary transformer.

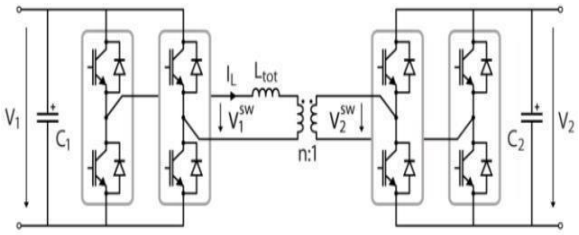


Fig. 2: Dual active Bridge Converter Topology

The topology of the dual active bridge converter is shown in the following figure. This system can be easily implemented using four PEB modules, or two PEH modules. Phase shifted full bridge (PSFB) DC-DC converters are used frequently to step down high DC bus voltages and/or provide isolation in medium to high power applications like server power supplies, telecom rectifiers, battery charging systems, and renewable energy.

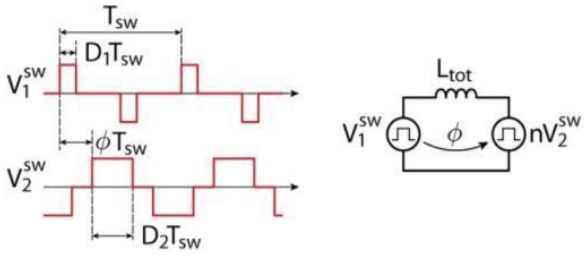


Fig. 3: Simplified loss-less scheme of the DAB converter

Electric Energy Storage Systems

Energy storage and high-voltage power electronics are key technologies transforming the electric power sector. Since electricity generation is relatively inflexible over short timeframes, storage systems help balance supply and demand, especially during peak loads. They also enable smoother integration of renewable energy, support microgrid stability, and enhance power reliability for sensitive industrial operations. Power electronics like switches and inverters allow precise, rapid control of electricity flow, improving efficiency and resilience in long-distance transmission. Reducing the cost and boosting the adoption of these technologies is a major focus for advancing the future energy grid.

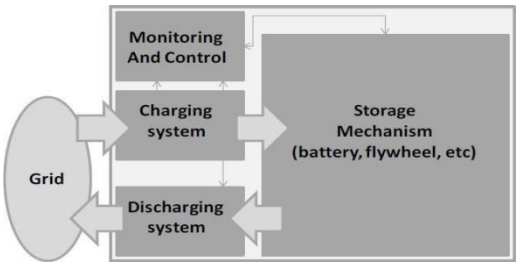


Fig. 4: Basic components of Energy Storage System

The charging mechanism in an energy storage system converts grid power into a form suitable for storage, such as using a motor to spin a flywheel or heating elements in thermal storage. Energy is later extracted through a discharge mechanism like reversing the motor to act as a generator in flywheels or using heat exchangers in thermal systems to supply electricity back to the grid. A critical component is the control system, which monitors and manages both charging and discharging processes, ensuring safe operation, efficient energy flow, and optimal performance based on system demands.

Electric Vehicle: Electric vehicles (EVs), powered by electric motors using energy from batteries, fuel cells, or external sources, have gained significant momentum in the 21st century due to advancements in technology and growing environmental concerns. Originally introduced in the mid-19th century, EVs offered superior comfort and ease of use compared to early gasoline vehicles, though internal combustion engines dominated for decades. Today, with increased focus on reducing greenhouse gas emissions, air pollution, and reliance on fossil fuels, EVs are viewed as a key solution to combat climate change. Government incentives, public awareness, and green recovery initiatives have accelerated EV adoption globally, particularly in regions like North America, Europe, and China. According to the International Energy Agency, EVs could reach a 30% global market share by 2030, underscoring their vital role in the future of sustainable transportation.

A PI Controller stands for Proportional-Integral Controller. It's a type of feedback controller used in control systems to maintain a process variable (like speed, voltage, temperature, etc.) at a desired set point.

It's one of the most commonly used control strategies in engineering due to its simplicity and effectiveness.

It has two parts:

1. Proportional (P) – Reacts to the current error

2. Integral (I) – Reacts to the accumulated error over time

The controller output is:

$u(t)=Kp\cdot e(t)+Ki\cdot\int e(t)\,dtu(t)$

Where:

- u(t) = output (control signal)
- e(t)= error (setpoint – actual value)
- Kp = proportional gain
- Ki= integral g

Table 1: Ziegler – Nichols Controller Settings

Type of Control	$G_c(s)$	K_c	T_I	T_D
Proportional	K_c	K_{cu}	-	-
Proportional – Integral	$\left(K_c+\frac{1}{T_IS}\right)$	0.45 K_{cu}	$\frac{T_u}{1-2}$	-
PID	$K_c\left(1+\frac{1}{T_IS}+T_DS\right)$	0.6 K_{cu}	$\frac{T_u}{2}$	$\frac{T_D}{8}$

The block diagram in Figure 5 illustrates the implementation of a PI controller in buck and boost converters. It compares the actual output voltage with a constant reference voltage to produce an error signal. This error

signal is amplified and fed into the PI controller, which generates a control signal to adjust the switching times of the converter's regulator switch. By doing so, the PI controller ensures a stable output voltage (Vo), effectively compensating for variations in input voltage and load conditions.

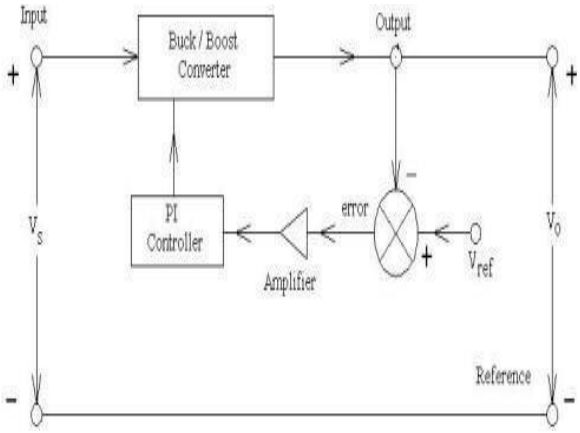


Fig. 5: Block Diagram of the Buck/ Boost Converter with the PI Controller

Artificial Neural Network (ANN): An Artificial Neural Network (ANN) controller is an intelligent control system that emulates the human brain's ability to learn and adapt. Comprising layers of interconnected neurons, it can identify patterns and relationships within data. ANN controllers are particularly useful in managing complex, non-linear, or time-varying systems that are challenging to model using conventional mathematical methods, making them ideal for advanced control applications.

1. Input Layer: Receives data from the system (like sensor values or error signals).
2. Hidden Layers: Process data using weighted connections and activation functions.
3. Output Layer: Produces control signals for actuators or the system being controlled.

The network is trained using data (often from simulations or real-world observations),

allowing it to learn how to respond to various system station

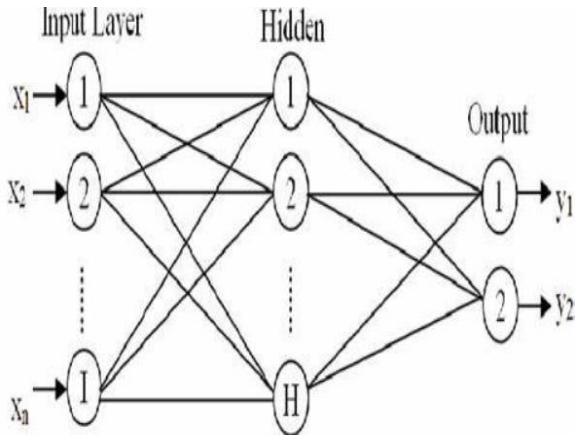


Fig. 6: ANN Structure

The ANN controller has two outputs one for voltage level control via the grid corrector and another for current control via the Buck/Boost conversion to maintain load current. Grid tension and DC-link tension control are used to control tension controlled system. As result, some parameters are estimated as critical inputs to controller. To begin, there is an error between DC estimated link's voltage and established reference voltage. Calculated error is then communicated to ANN control unit by establishing corresponding shift angle.Second, from immediate grid voltage, direct and quadrature voltages are generated. Real voltage, in comparison to desired direct voltage, an error that specifies amount of modulation signal required between two as

input to ANN controller for controlling DC/DC converter.

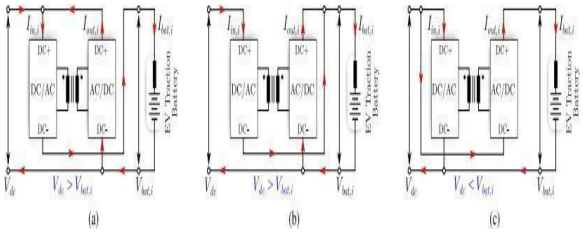


Fig. 7: Feasible partial power charger unit implementation schemes

Partial Power Processing Unit (PPPU): A PPCU is an advanced charging interface for electric vehicles that processes only a fraction of the total power, unlike conventional chargers that handle full power delivery. Most of the power flows directly from the DC link to the EV battery, while the PPCU regulates the charging rate as an active series-pass element. It ensures controlled charging without power loss, avoids circulating currents between chargers, and complies with IEC ripple standards. Typically realized using isolated two-port DC–DC converters, PPCUs allow efficient and flexible charging, though a universal connection scheme is still being explored.

IV. RESULTS

Simulation Results Only With PI Controller

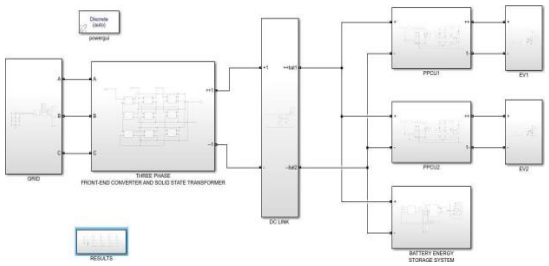


Fig. 8: Simulation Schematic Diagram

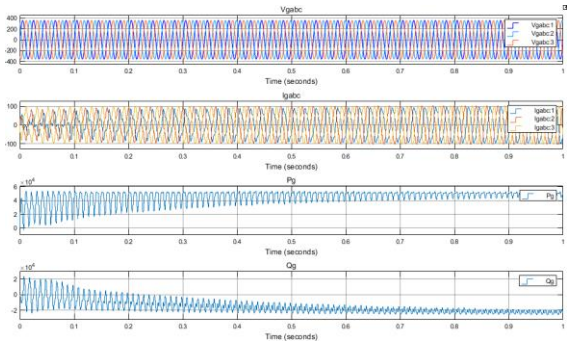


Fig. 9: Grid related waveform

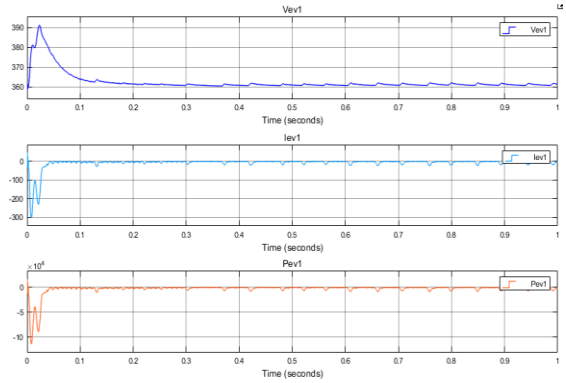


Fig. 10: EV-1 load related waveform

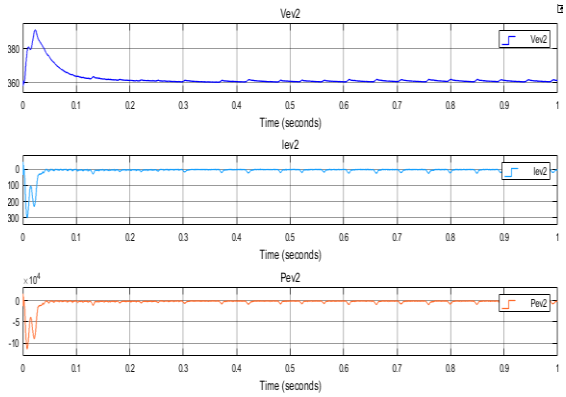


Fig. 11: EV-2 load related waveform

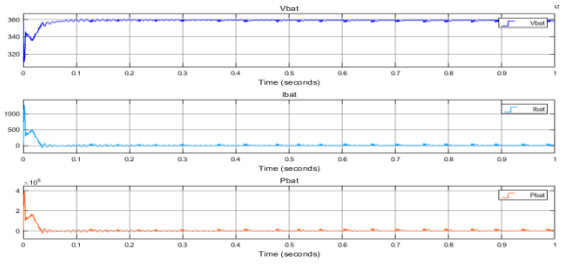


Fig. 12: DC Link related waveform

Simulation Results Using Fuzzy Logic Controller: ANN based Simulation Result

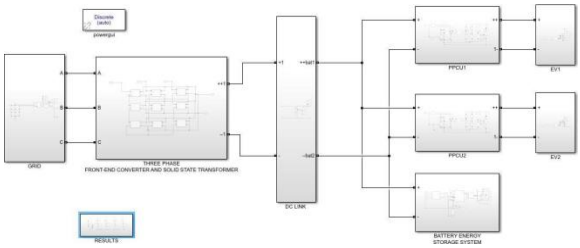


Fig. 13: Schematic diagram using ANN controller.

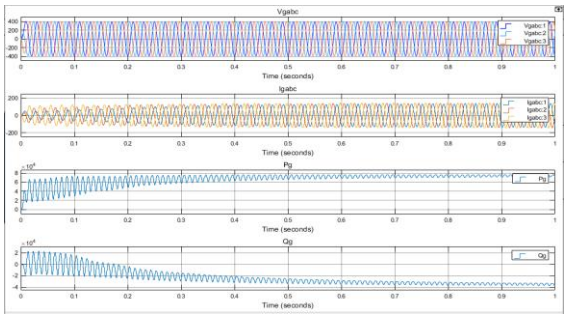


Fig. 14: Grid related waveform

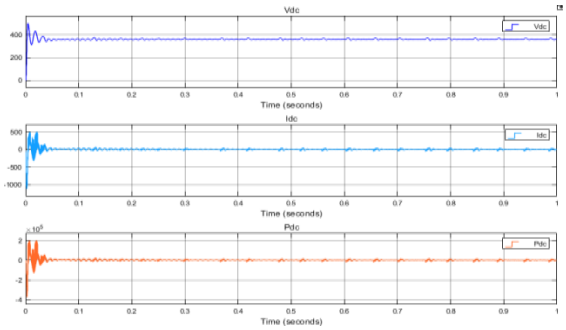


Fig. 16: DC Link related waveform

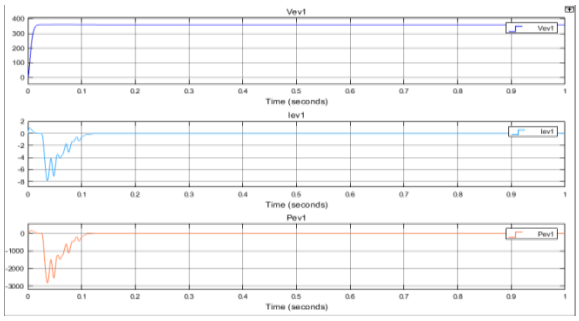


Fig. 17: EV-1 load related waveform

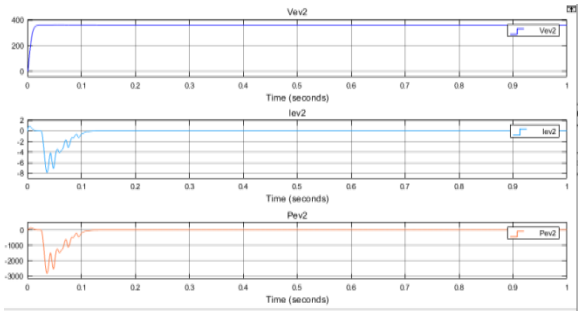


Fig. 18: EV-2 load related waveform

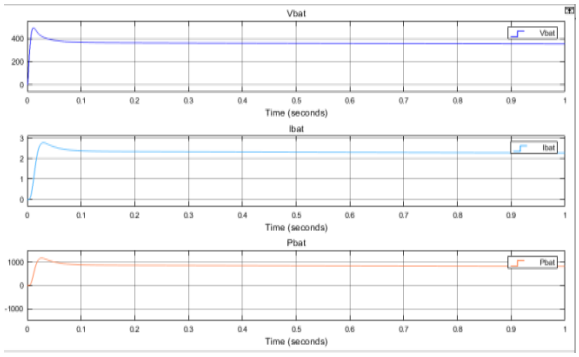


Fig. 19: Battery Storage System waveform using ANN controller

V. CONCLUSION

The proposed advanced charging station architecture, combining partial power processing with an Artificial Neural Network (ANN) controller, offers a highly efficient and intelligent solution for modern EV charging demands. By utilizing ANN-based control, the system ensures optimized power flow with reduced energy losses, dynamically adapts to varying grid and load conditions, and delivers improved voltage regulation and response times promoting battery health and charging safety. Its scalability and resilience in non-linear, uncertain environments further enhance its effectiveness. The ANN integration strengthens real-time decision-making and predictive control, outperforming traditional controllers in complex charging scenarios.

VI. REFERENCES

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